# Algebra 1 Worksheet

## Properties of Real Numbers, Simplifying Expressions, and Operations with Signed Numbers

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Name: _____ Date: _____

Class/Period: _____

## Part 1 — Quick Reference: Key Properties & Rules

### A. Properties of Addition

- **Commutative Property:** $a + b = b + a$ (you can add in any order)
  *Example:* $3 + 7 = 7 + 3 = 10$
- **Associative Property:** $(a + b) + c = a + (b + c)$ (you can regroup)
  *Example:* $(2 + 5) + 8 = 2 + (5 + 8) = 15$
- **Identity Property:** $a + 0 = a$ (adding 0 changes nothing)
  *Example:* $-9 + 0 = -9$
- **Inverse Property:** $a + (-a) = 0$ (opposites add to zero)
  *Example:* $6 + (-6) = 0$

### B. Properties of Multiplication

- **Commutative Property:** $a \cdot b = b \cdot a$ (you can multiply in any order)
  *Example:* $4 \times 9 = 9 \times 4 = 36$
- **Associative Property:** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ (you can regroup)
  *Example:* $(2 \times 3) \times 5 = 2 \times (3 \times 5) = 30$
- **Identity Property:** $a \cdot 1 = a$ (multiplying by 1 changes nothing)
  *Example:* $-7 \times 1 = -7$
- **Inverse Property:** $a \times (1/a) = 1$, where $a \neq 0$ (reciprocals multiply to 1)
  *Example:* $5 \times (1/5) = 1$
- **Zero Property:** $a \times 0 = 0$ (anything times zero is zero)
  *Example:* $-12 \times 0 = 0$

- **Distributive Property:** $a(b+c) = ab + ac$ (multiply through the parentheses)

  *Example:* $3(x+4) = 3x + 12$

### C. Simplifying Expressions Using Properties

Use the Distributive Property to clear parentheses, then combine like terms (using the Commutative and Associative Properties to reorder and regroup).

*Example:* $5(2x-3)+4 \;\rightarrow\; 10x - 15 + 4$ *(distribute)* $\;\rightarrow\; 10x - 11$ *(combine like terms)*

### D. Adding & Subtracting Real Numbers (including negatives, decimals, and fractions)

- Same signs: add the absolute values and keep the common sign.

  *Example:* $-4 + (-9) = -13$

- Different signs: subtract the smaller absolute value from the larger, and keep the sign of the number with the larger absolute value.

  *Example:* $-8 + 5 = -3$

- Subtracting a number = adding its opposite: $a - b = a + (-b)$

  *Example:* $7 - 12 = 7 + (-12) = -5$

- Decimals: line up the decimal points, then add/subtract as usual.

  *Example:* $-3.6 + 1.25 = -2.35$

- Fractions: rewrite with a common denominator, then add/subtract the numerators.

  *Example:* $3/4 + (-1/2) = 3/4 - 2/4 = 1/4$

## Part 2 — Multiple Choice (10 Questions: 3 Easy, 4 Medium, 3 Hard)

*Circle the best answer.*

**1.** Which property is shown below? ***[Easy]***
$5 + 8 = 8 + 5$
A) Associative Property of Addition
B) Commutative Property of Addition
C) Identity Property of Addition
D) Distributive Property

**2.** Simplify: $-7 + 3$ ***[Easy]***
A) $-10$ B) $10$ C) $-4$ D) $4$

**3.** Which property is shown below? ***[Easy]***
$6 \times 1 = 6$
A) Identity Property of Multiplication
B) Zero Property of Multiplication
C) Commutative Property of Multiplication
D) Inverse Property of Multiplication

**4.** Which property justifies the equation below? ***[Medium]***
$4(x + 5) = 4x + 20$
A) Associative Property
B) Commutative Property
C) Distributive Property
D) Identity Property

**5.** Simplify: $-3.5 - (-2.8)$ ***[Medium]***
A) $-6.3$ B) $-0.7$ C) $0.7$ D) $6.3$

**6.** Simplify: $\frac{2}{3} + \left(-\frac{5}{6}\right)$ ***[Medium]***
A) $-\frac{1}{6}$ B) $\frac{1}{6}$ C) $-\frac{3}{2}$ D) $\frac{3}{2}$

**7.** Simplify the expression using the properties of real numbers: $5(2x - 3) + 4$ ***[Medium]***
A) $10x - 11$ B) $10x + 11$ C) $7x - 11$ D) $2x - 11$

**8.** Simplify: $-4(2x - 5) - 3(x + 1)$ ***[Hard]***
A) $-11x + 17$ B) $-11x - 17$ C) $-5x + 17$ D) $11x + 17$

**9.** Simplify: $-\frac{3}{4} - 1\frac{5}{8}$ ***[Hard]***
A) $-2\frac{3}{8}$ B) $-1\frac{3}{8}$ C) $2\frac{3}{8}$ D) $-2\frac{5}{8}$

**10.** Which property is used to rewrite the expression below? ***[Hard]***
$6 + (4 + x) = (6 + 4) + x$
A) Associative Property of Addition
B) Commutative Property of Addition
C) Distributive Property

D) Identity Property of Addition

## Part 3 — Fill in the Blank (10 Questions)

*Write your answer in the blank, and show your work in the space provided.*

**1.** $-8 + _____ = 0$ ***[Easy]***
*Show your work:*

**2.** Simplify: $-5 + (-6) = _____$ ***[Easy]***
*Show your work:*

**3.** $9 + 4 = 4 + 9$ illustrates the _____ Property of Addition. ***[Easy]***
*Show your work:*

**4.** Simplify: $-2.4 - 3.9 = _____$ ***[Medium]***
*Show your work:*

**5.** Simplify: $-\frac{1}{4} + \frac{5}{8} = _____$ ***[Medium]***
*Show your work:*

**6.** Complete the equation using the Distributive Property: $6(x-2) =$ _____ *[Medium]*
*Show your work:*

**7.** Simplify the expression: $3(4x+1) - 5x =$ _____ *[Medium]*
*Show your work:*

**8.** Simplify: $-6\frac{1}{3} - 2\frac{3}{4} =$ _____ *[Hard]*
*Show your work:*

**9.** Simplify: $-5(2x-4) + 3(x-6) =$ _____ *[Hard]*
*Show your work:*

**10.** $2 \times (3 \times 7) = (2 \times 3) \times 7$ illustrates the _____ Property of Multiplication. *[Hard]*
*Show your work:*

## Part 4 — Word Problems (Apply the Skills Above)

**1.** The temperature was $-6°$F early in the morning. By noon, the temperature had risen $15°$F. What was the temperature at noon? ***[Easy]***

*Show your work:*

**2.** Maria is buying 4 notebooks. Each notebook normally costs \$3.25, but there is a \$0.50 discount on every notebook today. Use the Distributive Property to write an expression for the cost of the notebooks, then simplify it. She also buys one pack of pens for \$4.50 — what is her total cost? ***[Medium]***

*Show your work:*

**3.** A scuba diver starts at a depth of $18\frac{3}{4}$ feet below sea level ($-18\frac{3}{4}$ ft). She descends (goes down) another $12\frac{1}{2}$ feet, then ascends (comes up) $7\frac{1}{3}$ feet. Write an expression for her final position relative to sea level and simplify it. Name one property you could use to reorder the terms before combining them. ***[Hard]***

*Show your work:*

## Answer Key — Detailed Solutions

### Part 2 — Multiple Choice

**1. Answer: B — Commutative Property of Addition**

The equation only swaps the order of the two addends (5+8 becomes 8+5); the values themselves and the grouping don't change. Reordering addition is exactly the Commutative Property.

**2. Answer: C — $-4$**

$-7$ and 3 have different signs, so subtract the smaller absolute value from the larger: $|-7| = 7$, $|3| = 3$, and $7 - 3 = 4$. The sign of the answer matches $-7$ (the larger absolute value), so $-7 + 3 = -4$.

**3. Answer: A — Identity Property of Multiplication**

Multiplying any number by 1 leaves it unchanged: $6 \times 1 = 6$. This is the defining feature of the Identity Property of Multiplication (the Identity Property of Addition instead uses 0).

**4. Answer: C — Distributive Property**

$4(x + 5) = 4x + 20$ multiplies the 4 through both terms inside the parentheses ($4 \cdot x$ and $4 \cdot 5$). Multiplying a factor across a sum like this is the Distributive Property.

**5. Answer: B — $-0.7$**

Subtracting a negative is the same as adding its opposite: $-3.5 - (-2.8) = -3.5 + 2.8$. These have different signs, so subtract absolute values: $3.5 - 2.8 = 0.7$, and the sign follows $-3.5$ (larger absolute value), giving $-0.7$.

**6. Answer: A — $-\frac{1}{6}$**

Rewrite with a common denominator of 6: $\frac{2}{3} = \frac{4}{6}$. Then $\frac{4}{6} + \left(-\frac{5}{6}\right) = \frac{4-5}{6} = -\frac{1}{6}$.

**7. Answer: A — $10x - 11$**

Distribute: $5(2x - 3) = 10x - 15$. Then combine like terms with the $+4$: $10x - 15 + 4 = 10x - 11$.

**8. Answer: A — $-11x + 17$**

Distribute both terms: $-4(2x - 5) = -8x + 20$, and $-3(x + 1) = -3x - 3$. Combine: $(-8x - 3x) + (20 - 3) = -11x + 17$.

**9. Answer: A — $-2\frac{3}{8}$**

Write $1\frac{5}{8}$ as an improper fraction: $\frac{13}{8}$. Rewrite $-\frac{3}{4}$ with denominator 8: $-\frac{6}{8}$. Then $-\frac{6}{8} - \frac{13}{8} = -\frac{19}{8}$, which as a mixed number is $-2\frac{3}{8}$.

**10. Answer: A — Associative Property of Addition**

The three terms (6, 4, and $x$) stay in the same order the entire time — only the grouping (parentheses) moves from around $(4 + x)$ to around $(6 + 4)$. Regrouping without changing order is the Associative Property.

### Part 3 — Fill in the Blank

**1. Answer:** 8

The Inverse Property of Addition says a number plus its opposite equals 0: $a + (-a) = 0$. Since $-8$'s opposite is 8, the blank is 8: $-8 + 8 = 0$.

**2. Answer:** $-11$

$-5$ and $-6$ have the same sign, so add their absolute values ($5 + 6 = 11$) and keep the common sign (negative): $-5 + (-6) = -11$.

**3. Answer: Commutative**

The two addends simply switch places ($9 + 4$ becomes $4 + 9$) with no change in grouping or value — that is the Commutative Property of Addition.

**4. Answer:** $-6.3$

Both numbers are negative in effect ($-2.4$ and $-3.9$ is like adding $-2.4 + (-3.9)$), so add the absolute values ($2.4 + 3.9 = 6.3$) and keep the negative sign: $-2.4 - 3.9 = -6.3$.

**5. Answer:** $\dfrac{3}{8}$

Rewrite with a common denominator of 8: $-\dfrac{1}{4} = -\dfrac{2}{8}$. Then $-\dfrac{2}{8} + \dfrac{5}{8} = \dfrac{-2+5}{8} = \dfrac{3}{8}$.

**6. Answer:** $6x - 12$

Distribute the 6 through both terms inside the parentheses: $6(x - 2) = 6 \cdot x - 6 \cdot 2 = 6x - 12$.

**7. Answer:** $7x + 3$

Distribute: $3(4x + 1) = 12x + 3$. Then combine like terms with $-5x$: $12x + 3 - 5x = 7x + 3$.

**8. Answer:** $-9\dfrac{1}{12}$

Rewrite both fractions with denominator 12: $-6\dfrac{1}{3} = -6\dfrac{4}{12}$ and $-2\dfrac{3}{4} = -2\dfrac{9}{12}$. Add the whole numbers and fractions separately: $-6 - 2 = -8$, and $-\dfrac{4}{12} - \dfrac{9}{12} = -\dfrac{13}{12} = -1\dfrac{1}{12}$. Combine: $-8 - 1\dfrac{1}{12} = -9\dfrac{1}{12}$.

**9. Answer:** $-7x + 2$

Distribute both terms: $-5(2x - 4) = -10x + 20$, and $3(x - 6) = 3x - 18$. Combine: $(-10x + 3x) + (20 - 18) = -7x + 2$.

**10. Answer: Associative**

The three factors (2, 3, and 7) stay in the same order — only the grouping (parentheses) moves from around ($3 \times 7$) to around ($2 \times 3$). Regrouping without changing order is the Associative Property.

## Part 4 — Word Problems

**1. Answer:** $9°$**F**

Start at $-6°$F and add a rise of $15°$F: $-6 + 15$. These have different signs, so subtract absolute values ($15 - 6 = 9$) and keep the sign of the larger absolute value (positive, since $|15| > |-6|$). Final temperature: $-6 + 15 = 9°$F.

**2. Answer: Expression** $4(3.25 - 0.50)$**; Notebooks cost \$11.00; Total with pens = \$15.50**

Each notebook costs \$3.25 with a \$0.50 discount, so each one really costs $(3.25 - 0.50)$. For 4 notebooks: $4(3.25 - 0.50)$. Using the Distributive Property: $4(3.25) - 4(0.50) = 13.00 - 2.00 = 11.00$. Adding the \$4.50 pack of pens: $11.00 + 4.50 = \$15.50$.

**3. Answer: Expression** $-18\frac{3}{4} - 12\frac{1}{2} + 7\frac{1}{3}$**; Final position** $= -23\frac{11}{12}$ **ft** ($23\frac{11}{12}$ **feet below sea level)**

Convert all fractions to twelfths: $-18\frac{3}{4} = -18\frac{9}{12}$, $-12\frac{1}{2} = -12\frac{6}{12}$, and $7\frac{1}{3} = 7\frac{4}{12}$. You can use the Commutative Property of Addition to reorder the terms so the whole numbers and fractions are easier to combine: $(-18 - 12 + 7) + \left(-\frac{9}{12} - \frac{6}{12} + \frac{4}{12}\right) = -23 + \left(-\frac{11}{12}\right) = -23\frac{11}{12}$ ft.

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